Problem1 最大似然学习

公式 20.5，线性高斯模型，有连续父变量 X 和连续子变量 Y， Y 符合高斯分布，均值线性依赖 X 的值，标准差是固定的，我们要学习条件分布 P(y|x)，最大化条件似然

P(y|x) = $\frac{1}{\sqrt{2\pi}\sigma}e^{\frac{(y-(\theta_1 x+\theta_2))^2}{2\sigma^2}}$然后每个数据累加. 取 log 可以得到

$$\mathrm{L} = \sum_{j=1}^{N} log\frac{1}{\sqrt{2\pi}\sigma}e^{\frac{(y-(\theta_1 x+\theta_2))^2}{2\sigma^2}} = -\mathrm{N}\left(\log\sigma + \log\sqrt{2\pi}\right) - \sum_j \frac{\left(y_j - \left(\theta_1 x_j + \theta_2\right)\right)^2}{2\sigma^2}$$

令导数为 0 得到

$$\frac{\partial L}{\partial \theta_1} = -\sum_j \frac{x_j(y_j - (\theta_1 x_j + \theta_2))}{\sigma^2} = 0$$

$$\frac{\partial L}{\partial \theta_2} = -\sum_j \frac{(y_j - (\theta_1 x_j + \theta_2))}{\sigma^2} = 0$$

$$\frac{\partial L}{\partial \sigma} = -\frac{N}{\sigma} + \sum_j \frac{\left(y_j - \left(\theta_1 x_j + \theta_2\right)\right)^2}{\sigma^3} = 0$$

我们容易得到

$$\sum_j \frac{(y_j - (\theta_1 x_j + \theta_2))}{1} = 0 \text{ 这个可以求出 theta2 和 theta1 的关系}$$

$$\sigma^2 > 0$$

$$\sum_j \frac{\left(y_j - \left(\theta_1 x_j + \theta_2\right)\right)^2}{\sigma^2} = \mathrm{N} \text{ 这个可以求出}\sigma^2$$

然后来求 theta1

$$\sum_j \frac{x_j(y_j - (\theta_1 x_j + \frac{1}{N}\sum_j(y_j - \theta_1 x_j)))}{1} = 0$$

$$\sum_j x_j\left(y_j - \left(\theta_1 x_j + \frac{1}{N}\sum_j\left(y_j - \theta_1 x_j\right)\right)\right) = \sum_j x_j y_j - \theta_1 x_j^2 - \frac{x_j}{N}\sum_j\left(y_j - \theta_1 x_j\right) = 0$$

这里化简过程就不打字了

可以知道参数是:

$$\theta_1 = \frac{N\left(\sum_j x_j y_j\right) - \left(\sum_j y_j\right)\left(\sum_j x_j\right)}{N\left(\sum_j(x_j^2)\right) - \left(\sum_j x_j\right)^2}$$

$$\theta_2 = \frac{1}{N}\sum_j(y_j - \theta_1 x_j)$$

$$\sigma^2 = \frac{1}{N}\sum_j \left(y_j - \left(\theta_1 x_j + \theta_2\right)\right)^2$$

Problem2:

寻找 前向概率,

$$\mathrm{P(Y = true) = \pi}$$

a. 写 D 的 likelihood 的表达式,

$$\mathrm{D} = \pi^p (1-\pi)^n$$

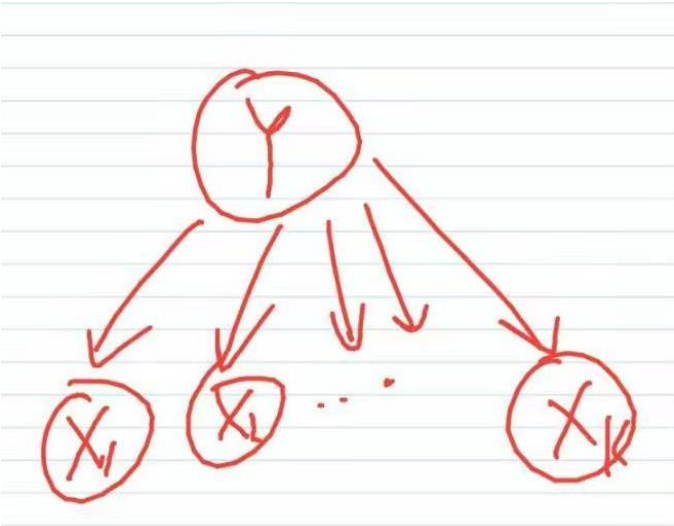
b.

$$\mathrm{L} = \ \mathrm{logD} = \mathrm{plog\pi + n\ \ log(1-\pi)}$$

$$\frac{\partial L}{\partial \pi} = \ \frac{p}{\pi} - \frac{n}{1-\pi} = 0 \ \textit{可以得到}\ \pi = \frac{\mathrm{p}}{p+n}$$

c.

求贝叶斯网络 .



Xi 是 attributes 也就是属性变量, 是叶结点, 需要预测的类变量 Y 是根结点. 假设各个属性彼此是条件独立的.

d.

$\mathrm{Xi = true\ , Y = true}$, the possibility $= \pi * \alpha_i$, *the number of this case* $= \ p_i^+$

$\mathrm{Xi = true\ , Y = false}$, the possibility $= \ (1-\pi) * \beta_i$, *the number of this case* $= \ p_i^-$

$\mathrm{Xi = false\ , Y = true}$, the possibility $= \pi * (1-\alpha_i)$, *the number of this case* $= \ n_i^+$

$\mathrm{Xi = false\ , Y = false}$, the possibility $= \ (1-\pi) * (1-\beta_i)$, *the number of this case* $= \ n_i^-$

$$\left((1-\pi)^{n_i^-} * (1-\beta_i)^{n_i^-}\right) * (\pi * (1-\alpha_i))^{n_i^+} * \left((1-\pi) * \beta_i\right)^{p_i^-} \ * (\ \pi * \alpha_i)^{p_i^+}$$

$$\pi^p(1-\pi)^n \prod_i (1-\beta_i)\hat{}n_i^- \ \beta_i\hat{}p_i^- \ (1-\alpha_i)\hat{}n_i^+ \ \alpha_i\hat{} \ p_i^+$$

e.

L = $\mathrm{plog}\pi + \mathrm{n}\ \log(1-\pi) + \ \sum_i \ n_i^-\log(1-\beta_i) + n_i^+\log(1-\alpha_i) + p_i^+ \ log\alpha_i + p_i^- \ \log\beta_i$

$$\frac{\partial L}{\partial \beta_i} = \frac{p_i^-}{\beta_i} - \frac{n_i^-}{1-\beta_i} = \ 0$$

$$\frac{\partial L}{\partial \alpha_i} = \frac{p_i^+}{\alpha_i} - \frac{n_i^+}{1-\alpha_i} = 0$$

$$\alpha_i = \frac{p_i^+}{p_i^+ + \ n_i^+}$$

$given$ $\mathrm{Y} = \mathrm{true}$, we could know P(X = true) =α_i , P(X= false) = 1-α_i

$$\beta_i = \frac{p_i^-}{p_i^- + n_i^-}$$

$given$ $\mathrm{Y} = \mathrm{false}$, we could know P(X = true) =β_i , P(X= false) =$1-\beta_i$